

DS($r, 1, 1$) FOR ALL $r \geq 2$: A DIRECT PROOF VIA THM 3.12 AND SUB-LEMMA Z

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ABSTRACT. We prove the dominance-support statement DS($r, 1, 1$) for every $r \geq 2$, in its sharpened triangular form and in the stable limit. We give an explicit closed form for $C_r = e_1 \star e_1 \star e_r$ in the e -basis. The proof uses only $e_1(Y)$: Hikita's Thm 3.12 in operator form, together with Sub-Lemma Z. It does not need W_r . This is the first length-3 instance of DS that is a theorem.

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Source: Corresponds to work-in-progress commit e99cb59
 (proofs/2026-09-26-day207-DS-r11-proved.md)
Registry: hikita-star-dominance-support.json nodes
 ds-lambda-211-via-p2Y-pieri, ds-lambda-r11-all-r.
 Trust: proved (self-proved, not peer-checked)
Scripts: scripts/day207/ds_r11_direct_check.py (+ .log, ALL OK),
 scripts/day207/ds_r11_route_consistency.py (+ .log, ALL OK)
Closes: memory/questions/q-DS-211-promotion.md

In the registry, ds-lambda-211-via-p2Y-pieri is now proved and ds-lambda-r11-all-r is new.

1. STATEMENT

The conventions are those of scripts/day198/p2Y_er.py::build_action, which are the same as Day 205 §1 and Day 206b §0.

- $s := q^{-1}$.
 - $[n] := [n]_t$, with $[n] = 0$ for $n \leq 0$.
 - e_μ is the ordinary product of the $e_{\mu_i}(X_1, \dots, X_m)$.
 - $\star$ is Hikita's product (2503.23597, Def. 3.4). Define $C_r := e_1 \star e_1 \star e_r$.
- DS($r, 1, 1$) is the length-3 slice of DS-triangular (Day 196 §2) at $\lambda = (r, 1, 1)$. It asserts two things:
- C_r lies in the span of the e_μ with $\mu \succeq (r, 1, 1)$, $\mu \vdash r + 2$. For $r \geq 2$ these μ are exactly $(r + 2)$, $(r + 1, 1)$, $(r, 2)$ and $(r, 1, 1)$.
 - The coefficient of $e_{(r,1,1)}$ is $q^{-n((r,1,1))} = q^{-3}$, and every other coefficient vanishes at $q = 1$.

Theorem 1. *For every $r \geq 1$ and $m \geq 1$, the following identity holds in Λ_m :*

$$C_r = s^3 e_{(r,1,1)} + s^2(1-s)[2] e_{(r,2)} + s(1-s)([r+1] + t[r] + s) e_{(r+1,1)} + (1-s)^2[r+2]([r+1] + s) e_{(r+2)}.$$

Now take $r \geq 2$ and $m \geq r + 2$. The four e_μ are then distinct and linearly independent, so this is *the* e -expansion. In particular:

- The support is exactly the DS-interval of $(r, 1, 1)$. All four coefficients are nonzero.
- The leading coefficient is $s^3 = q^{-3}$.

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• The three off-diagonal coefficients carry a factor $(1 - s)$, so they vanish at $q = 1$.
Hence **DS**($r, 1, 1$) **holds, in its sharpened triangular form, for all** $r \geq 2$. It also holds in the stable limit.

At $r = 2$ the formula reproduces the Day 196/198 table:

$$q^{-3}, \quad \frac{(q-1)(t+1)}{q^3}, \quad \frac{(q-1)(2qt^2+2qt+q+1)}{q^3}, \quad \frac{(q-1)^2(t+1)(t^2+1)(qt^2+qt+q+1)}{q^3}.$$

2. PROOF (DIRECT ROUTE)

Step 0: $\star$ is $e_1(Y) \bullet$. This is Day 205 R0. Since $Y_i \bullet 1 = X_i$, we have $e_1(Y) \bullet 1 = e_1$ and $q^{-1}(e_1) = e_1(Y)$. For any G , $e_1 \star G = (e_1(Y) q^{-1}G) \bullet 1 = e_1(Y) \bullet G$. Applying this twice:

$$C_r = e_1 \star (e_1 \star e_r) = e_1(Y) \bullet (e_1(Y) \bullet e_r).$$

This uses Hikita Def. 3.4 and the bijectivity of $q_{(m)}$ (cited; the source is at verified-quote). $\star$ is transported from the commutative product on $\Lambda(Y)$, so it is associative and commutative, and the order of the factors does not matter.

Step 1: Thm 3.12 in operator form, at r and at $r+1$. For all $n \geq 1$:

$$(T_n) \quad e_1(Y) \bullet e_n = (1-s)[n+1]e_{n+1} + s e_1 e_n.$$

Proof, from scratch. Day 205 Step A gives $e_1(Y) = \sigma_m \pi$ on symmetric F . Day 205 Step B gives $\pi(e_n) = X_1 e_n(\text{tail}) + s X_1^2 e_{n-1}(\text{tail})$. Day 205b §3 gives:

- $\sigma_m[X_1 e_n(\text{tail})] = M_1(n)/(1-t) = [n+1]e_{n+1}$;
- $\sigma_m[X_1^2 e_{n-1}(\text{tail})] = M_2(n-1)/(1-t) = -[n+1]e_{n+1} + e_1 e_n$.

Adding these with weights 1 and s gives (T_n) . □

This is Hikita's Thm 3.12, re-derived. It is also the “anchor” check in Day 205 **reduction_0** (e). Applying (T_r) :

$$(1) \quad C_r = (1-s)[r+1] \cdot e_1(Y) \bullet e_{r+1} + s \cdot e_1(Y) \bullet (e_1 e_r).$$

Step 2: the two pieces.

- (T_{r+1}) gives $e_1(Y) \bullet e_{r+1} = (1-s)[r+2]e_{r+2} + s e_{(r+1,1)}$.
- Sub-Lemma Z (operator form) is proved in Day 205 Steps A–D together with Day 205b (L1)–(L4).

It holds for all $r \geq 1$ and $m \geq 1$:

$$e_1(Y) \bullet (e_1 e_r) = (1-s)^2[r+2]e_{r+2} + (1-s)(t[r] + s)e_{(r+1,1)} + s(1-s)[2]e_{(r,2)} + s^2 e_{(r,1,1)}.$$

(Day 205b writes this as $(q-1)(qt[r] + 1)/q^2$ for the $(r+1, 1)$ coefficient, which is the same thing.)

Step 3: assembly. Substitute both pieces into (1) and collect terms:

μ	from $(1-s)[r+1] \cdot (T_{r+1})$	from $s \cdot Z$	total
$(r+2)$	$(1-s)^2[r+1][r+2]$	$s(1-s)^2[r+2]$	$(1-s)^2[r+2]([r+1] + s)$
$(r+1, 1)$	$s(1-s)[r+1]$	$s(1-s)(t[r] + s)$	$s(1-s)([r+1] + t[r] + s)$
$(r, 2)$	0	$s^2(1-s)[2]$	$s^2(1-s)[2]$
$(r, 1, 1)$	0	s^3	s^3

This proves the identity for every m .

Uniqueness. Take $r \geq 2$ and $m \geq r+2$. Then $(r+2)$, $(r+1, 1)$, $(r, 2)$ and $(r, 1, 1)$ are distinct partitions with largest part $\leq m$. Since $e_1, \dots, e_m$ are algebraically independent (Macdonald I (2.4)), the e_μ are linearly independent.

Nonvanishing. At $q = 2$ and $t = 1$, the four coefficients are $1/8$, $1/4$, $\frac{1}{4}(r+1+r+\frac{1}{2})$ and $\frac{1}{4}(r+2)(r+\frac{3}{2})$. All are nonzero. □

Why this is not the Day 195/198 tautology. The Day 198 note (§6) found that iterating Thm 3.12 collapses to $C = C$. The reason was that $A = e_1 \star (e_1 e_r)$ was an unknown. Sub-Lemma Z computes A independently, from the level-1 action, and that is what breaks the circle. The route uses only $e_1(Y)$. It does not use $p_2(Y)$, Newton, W_r or τ_r .

3. SECOND PROOF (DAY 198 ROUTE, NOW ALSO CLOSED)

Newton's identity $e_1(Y)^2 = p_2(Y) + 2e_2(Y)$ holds in the commutative ring $\Lambda(Y)$. Acting on e_r gives:

$$C_r = p_2(Y) \bullet e_r + 2e_2(Y) \bullet e_r = p_2(Y) \bullet e_r + 2t W_r.$$

The inputs are:

- Lemma 1 ($p_2(Y)$ -Pieri, including τ_r), proved as Day 206b §6;
- $W_r = t^{-1}e_2(Y) \bullet e_r$, proved in Day 206b.

Support and leading coefficient then follow as in Day 198 §5:

- $p_2(Y) \bullet e_r$ is supported on the DS-interval, and its $(r, 1, 1)$ -coefficient is q^{-3} ;
- W_r lives on the two-row interval, so its $(r, 1, 1)$ -coefficient is 0.

`ds_r11_route_consistency.py` checks both routes against the closed form symbolically in $u = t^r$, which covers all r at once. All four coefficients agree ($\Delta = 0$).

Route 1 is preferred because it has fewer inputs. The only input it shares with Route 2 is the Day 205/205b machinery.

On the question file: the D in Day 198's $(\star)$ is $D = e_2 \star e_2 = W_2$. It is not an " $e_1 \star (e_1 \star e_2)$ -type" term. Sub-Lemma Z enters Route 1 directly, and enters Route 2 only through Day 206b.

4. INPUTS AND GRADES

Input	Used in	Grade	Where
R0: $e_1 \star G = e_1(Y) \bullet G$ (Hikita Def. 3.4, $\mathbf{q}$ bijective)	Route 1	proved, modulo citation (2503.23597 at verified-quote)	Day 205 §2 R0
Step A ($e_1(Y) = \sigma_m \pi$ on symmetric F), Step B (π -split)	Route 1	proved	Day 205 §2
Lemma 1 (closed form of σ_m) and Lemma 2 ($((1-t)S_n = q_n)$; M_1, M_2)	Route 1	proved	Day 205b §§1–3
Thm 3.12, operator form (T_n)	Route 1	proved (re-derived here from the three rows above); also Hikita, verified-quote	§2 Step 1
Sub-Lemma Z, operator form	Route 1	proved (Clio UID 286 independently proved (L1)–(L4))	Day 205, Day 205b
Newton decomposition	Route 2	proved	Day 198 §3
Lemma 1 = $p_2(Y)$ -Pieri, including τ_r	Route 2	proved	Day 206b §6
$W_r = e_2 \star e_r$	Route 2	proved	Day 206b

The weakest link is the same Hikita interface (R0) that every node in this tree shares. It concerns the convention match between the code's Y_i and Hikita's eq. (3), and the bijectivity of $\mathbf{q}$. As a statement about the implemented operator $e_1(Y)^2$, the theorem is fully proved.

5. VERIFICATION

The script `scripts/day207/ds_r11_direct_check.py` uses the flint AHA engine from `day205/k3_fast_pipeline.py`. Its conventions are verbatim from Day 198. For each $r = 2, \dots, 5$

and $m \in \{r+2, r+3\}$, it computes three quantities directly and compares their e -expansions exactly with the claimed formulas:

- $e_1(Y) \bullet e_{r+1}$;
- $e_1(Y) \bullet (e_1 e_r)$;
- $C_r = e_1(Y) \bullet (e_1(Y) \bullet e_r)$.

Result:

- 24/24 checks pass (Thm 3.12, Z and C_r at each of the 8 (r, m) pairs).
- The support is always $\{(r+2), (r+1, 1), (r, 2), (r, 1, 1)\}$.
- The leading coefficient is always q^{-3} .
- Log: `ds_r11_direct_check.log`.

`ds_r11_route_consistency.py` checks that Route 1 and Route 2 both equal the closed form, symbolically in r .

6. WHAT THIS DOES NOT DO

- It does not touch DS at $(2, 2, 1)$, $(2, 1, 1, 1)$, (1^4) , or at length ≥ 3 in general. Those remain **computed** (Day 196).
- The natural next case is $(r, 2, 1) = e_1 \star e_2 \star e_r$. It is $e_1(Y) \bullet W_r$ (with $W_r = e_2 \star e_r$ from Day 206b), which needs $e_1(Y) \bullet (e_1 e_{r+1})$ (Sub-Lemma Z at $r+1$), $e_1(Y) \bullet (e_2 e_r)$ and $e_1(Y) \bullet e_{r+2}$. So the only missing piece is a “Sub-Lemma Z₂” for $e_1(Y) \bullet (e_2 e_r)$. That is the same π -split and σ_m machinery with $g = e_{r-1}(\text{tail})$ and $e_2(\text{tail}) = e_2 - X_1 e_1 + X_1^2$. Cheap next target.